

NOTES 19: CLT-BASED INFERENCE FOR PROPORTIONS

Stat 120 | Spring 2026

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Name	Type of Variable(s)	Statistic	Parameter
Mean	Quantitative	$\bar{x}$	μ
Proportion	Categorical	$\hat{p}$	p
Standard Deviation	Quantitative	s	σ
Difference in Proportions	2 Categorical (1 Response, 1 Explanatory)	$\hat{p}_1 - \hat{p}_2$	$p_1 - p_2$
Difference in Means	1 Quantitative (Response) 1 Categorical (Explanatory)	$\bar{x}_1 - \bar{x}_2$	$\mu_1 - \mu_2$
Correlation	2 Quantitative	r	ρ
Slope	2 Quantitative	b_1	β_1

CLT: The Central Limit Theorem (CLT) tells us that if the sample size is big enough and the sample is random,

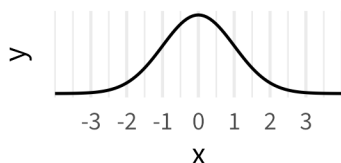
$$\bar{X} \sim N(\text{____}, \text{____})$$

$$\hat{p} \sim N(\text{____}, \text{____})$$

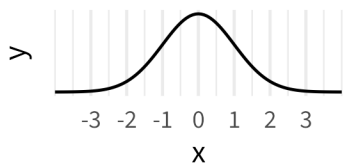
The general form for a **confidence interval** is:

Example: Finding z^*

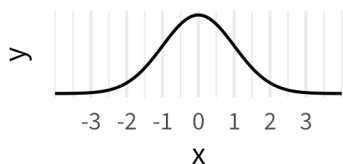
- 95% confidence interval



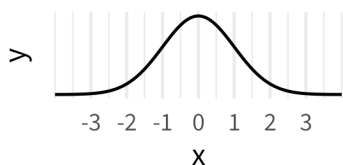
- 68% confidence interval



- 99% confidence interval



- 90% confidence interval



0.1 How big is big enough?

Example 1: $\hat{p} = .5$

Example 1: $\hat{p} = .05$

Rule of Thumb for Proportions:

- Expected count in each category (Yes/No) should be $> \rule{1cm}{0.4pt}$
- $np > \rule{1cm}{0.4pt}$ and $n(1 - p) > \rule{1cm}{0.4pt}$

0.2 How do we find the SE?

Standard Error for Proportions:

Idea: As n gets bigger, the SE gets $\rule{1cm}{0.4pt}$.

If $\hat{p}$ is close to .5, the SE is $\rule{1cm}{0.4pt}$ than if $\hat{p}$ is close to 0 or 1

Example: ESP Example Again

n = 14

```
p_hat = 3/14
p_0 = .2
n = 14
SE_p = sqrt((p_0*(1-p_0))/n)
SE_p
```

[1] 0.1069

```
z_score = (p_hat - .2)/SE_p
z_score
```

[1] 0.1336

```
p_val = pnorm(z_score, lower.tail = FALSE)
p_val
```

[1] 0.4468

n=1400:

n=14000:

Big Picture Picture

1 In-class Group Questions

1. APM Research Lab ran a [survey of likely Minnesota voters](#) between Sept 16-18 of 2024. They reported 48.4% in support of the Harris/Walz ticket, and 43.3% in support of the Trump/Vance ticket.

Here's an excerpt from their methodology report:

The margin for error, according to standards customarily used by statisticians, is no more than ± 3.5 percentage points. This means that there is a 95% probability that the "true" figure would fall within that range if all voters were surveyed.

- (a) Show how the margin of error was computed
 - (b) Is it OK to use the normal distribution as a model for the sampling distribution of a proportion in this case?
2. In a random sample of 100 moviegoers in January 2013, 64 of them said they are more likely to wait and watch a new movie in the comfort of their own home.
 - (a) Is it ok to use the normal distribution as a model for the sampling distribution of the proportion of moviegoers preferring to wait and watch at home?
 - (b) Suppose I want to know if more than 50% of moviegoers in the population prefer to wait and watch at home. Write down the corresponding null and alternative hypotheses.
 - (c) For the hypothesis test in part (b), find the standardized test statistic (i.e., the Z-score).

(d) Calculate the p-value and state a conclusion in context

(e) The table below gives some of the percentiles of the $N(0,1)$ distribution. If you were given this table (for example on an exam, where you do not have access to R), what could you say about the value of the p-value? (*Hint*: draw the picture)

percentage	percentile (qnorm(percentage))
90%	1.3
95%	1.6
97.5%	2.0
99%	2.3
99.5%	2.6

(f) Compute and interpret a 90% confidence interval for the population proportion of moviegoers who believe they are more likely to watch a new movie from home. (Use the information from the previous table to find the critical value needed.)

(g) How big of a sample size would you need if you wanted the margin of error for a 90% confidence interval to be no more than 2%? To answer this question, assume that the true proportion is $p = 0.5$, which would result in the biggest margin of error for a given sample size.